

ECON 3350/7350 Applied Econometrics for Macroeconomics and Finance

Lecture 5: Stochastic and Deterministic Trends

Fu Ouyang

School of Economics
University of Queensland

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Stochastic and Deterministic Trends

Recommended readings

Author	Title	Chapter	Call No
Enders	Applied Econometric Time Series, 4e	4	HB139 .E55 2015
Verbeek	A Guide to Modern Econometrics	8.2-8.5	HB139 .V465 2012

Trends

Many (not all) economic time series appear to have a trend in the mean and/or variance.

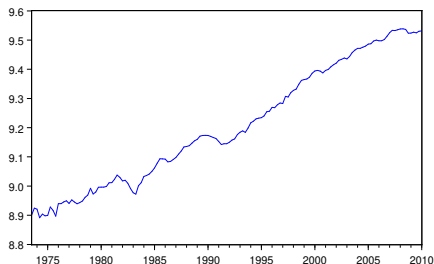


Figure 1: Australia log real GDP per capita.

What **type of trend** is present in this data, and how can we **transform it to stationary** data?

Trend Stationary

A trend can be modeled as **deterministic**, that is a **known function** (up to estimated parameters) of t . This is consistent with **exogenous** growth theories.

For example, a **linear trend** is specified by

$$y_t = a_0 + \delta t + a_1 y_{t-1} + \varepsilon_t, \quad |a_1| < 1.$$

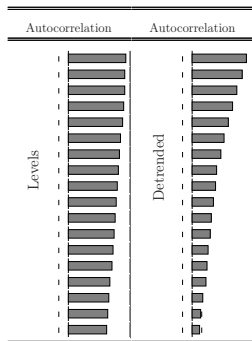
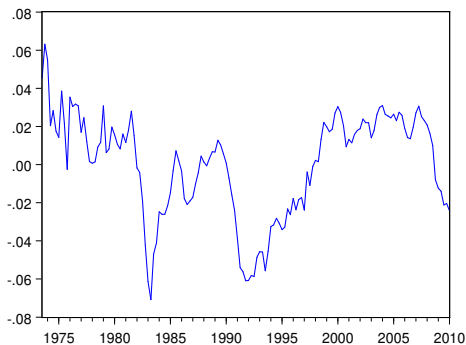
The trend can be “taken out” by setting $z_t = y_t - \delta/(1 - a_1)t$, which implies

$$z_t = \tilde{a}_0 + a_1 z_{t-1} + \varepsilon_t, \quad |a_1| < 1.$$

z_t is a **stationary** AR(1) process; $\delta/(1 - a_1)$ is the **long-run growth rate**.

De-trended Data

In practice, remove a linear trend by regressing y_t on a constant and t , then save the residuals (this will also de-mean the data).



Difference Stationary

Real Business Cycle theories suggestion innovations (i.e. productivity shocks) have a permanent effect on productivity.

Endogenous growth theories imply an alternative form of trend, e.g. arising from $\delta = 0$ and $a_1 = 1$:

$$y_t = a_0 + y_{t-1} + \varepsilon_t.$$

In this setting, the data contains a **stochastic trend**, where the expected change is $E(\Delta y_t) = a_0$; hence, a_0 is the **long-run growth rate**.

A stochastic trend is removed simply by **differencing** the data.

Difference Operator

The **difference operator** is defined in terms of the lag operator as $\Delta = 1 - L$. Hence,

$$\Delta y_t = (1 - L)y_t = y_t - y_{t-1}.$$

As with lag operators in general, we can combine two difference operator to construct a **difference-in-difference** operator, i.e.

$$\begin{aligned}\Delta^2 y_t &= \Delta(\Delta y_t) = (1 - L)(y_t - y_{t-1}) \\ &= (y_t - y_{t-1}) - (y_{t-1} - y_{t-2}) = \Delta y_t - \Delta y_{t-1}.\end{aligned}$$

In general,

$$\Delta^d = (1 - L)^d y_t = (1 - L)((1 - L)^{d-1} y_t) = \Delta^{d-1} y_t - \Delta^{d-1} y_{t-1}.$$

Difference Operator

Other notation you might come across:

$$\Delta_1 y_t = (1 - L)y_t = y_t - y_{t-1} = \Delta y_t,$$

$$\Delta_2 y_t = (1 - L^2)y_t = y_t - y_{t-2} \neq \Delta^2 y_t,$$

$$\vdots$$

$$\Delta_s y_t = (1 - L^s)y_t = y_t - y_{t-s} \neq \Delta^s y_t.$$

Finally, we can mix these to obtain:

$$\Delta_s^d = (1 - L^s)^d y_t = \Delta_s \cdots \Delta_s y_t \text{ i.e., applied } d \text{ times.}$$

Stochastic Trends

Recall that in the **stationary** AR(1) with

$$\begin{aligned}y_t &= a_0 + a_1 y_{t-1} + \varepsilon_t, \quad |a_1| < 1, \\&= a_0(1 + a_1 + \cdots + a_1^{t-1}) + a_1^t y_0 + \varepsilon_t + a_1 \varepsilon_{t-1} + \cdots + a_1^{t-1} \varepsilon_1,\end{aligned}$$

the effect of a shock dies out as t increases.

In the presence of a **stochastic trend** characterized by $a_1 = 1$,

$$\begin{aligned}y_t &= a_0 + y_{t-1} + \varepsilon_t, \\&= a_0 t + y_0 + \varepsilon_t + \varepsilon_{t-1} + \cdots + \varepsilon_1,\end{aligned}$$

all past shocks have **permanent** effects.

The AR(1) with $a_0 = 0$ and $a_1 = 1$ is called a **random walk** (RW).

- Intuitively, starting from y_0 and for large enough t , the random variable y_t can attain *any* value with equal probability.

Properties of the Random Walk

The RW does not have a finite **unconditional** expectation, but it does have the following **conditional** expectations:

$$E(y_t | y_{t-1}) = y_{t-1}, \quad E(y_t | y_0) = y_0.$$

Therefore, the mean is constant conditional on a particular initial value y_0 .

Conditional on an initial value y_0 , the variance for a RW is:

$$E((y_t - y_0)^2 | y_0) = \text{Var}(\varepsilon_t + \cdots + \varepsilon_1 | y_0) = t\sigma_\varepsilon^2,$$

and the covariance for observations s periods apart is:

$$E((y_t - y_0)(y_{t-s} - y_0) | y_0) = (t - s)\sigma_\varepsilon^2.$$

The variance and covariances are **not constant** $\Rightarrow$ the RW is **not stationary**.

Random Walk with Drift

The AR(1) with $a_0 \neq 0$ and $a_1 = 1$ is called a **random walk with a drift** (RW).

The sign of a_0 indicates the direction that y_t will drift over time; however, the actual trend is influenced by the shocks in each period.

The moments conditional on the initial value y_0 are:

$$E(y_t | y_0) = a_0 t + y_0,$$

$$Var(y_t | y_0) = Var(a_0 t + y_0 + \varepsilon_t + \cdots + \varepsilon_1 | y_0) = t\sigma_\varepsilon^2,$$

$$Cov(y_t, y_{t-s} | y_0) = (t-s)\sigma_\varepsilon^2.$$

Therefore, all moments depend on t .

Characteristic Roots

For more general processes with $p > 1$, stationarity is captured by the **characteristic equation**:

$$a(z) = 1 - a_1z - \cdots - a_pz^p = 0. \quad (1)$$

The roots of $a(z)$ —that is, all values of z that solve (1)—are called the **characteristic roots** of the system.

The **characteristic roots** control the **dynamic responses** of y_t to any shock.

- If *all* $|z| > 1$, then the effect of a shock **vanishes** over time.
- If there exists *at least one* $|z| = 1$, then the effect of a shock **persists** over time.
- If there exists *at least one* $|z| < 1$, then the effect of a shock **increases** over time.

Unit Root Processes

For the AR(1), the characteristic equation is $a(z) = 1 - a_1z = 0$, and the only root is $z = 1/a_1$.

- $a_1 = 1$ corresponds to $z = 1$, which we call the **unit root**.

For the AR(p), the lag polynomial can be factored as

$$a(L) = (1 - \alpha_1 L)(1 - \alpha_2 L) \cdots (1 - \alpha_p L),$$

where $\alpha_1, \dots, \alpha_p$ are functions of $a_1, \dots, a_p$.

If any $\alpha_j = 1$ for $j = 1, \dots, p$, then the process will have a **unit root**. If there are d such $\alpha_j = 1$, then the process will have d unit roots.

AR(2) Example

For the model

$$y_t = a_1 y_{t-1} + a_2 y_{t-2} + \varepsilon_t,$$

the characteristic polynomial is $a(z) = 1 - a_1 z - a_2 z^2 = 0$.

Completing the square (or using the quadratic formula), we can factor $a(L)$ as

$$a(L) = \left(1 + \frac{2a_1}{a_2 + \sqrt{a_2^2 + 4a_1}}L\right) \left(1 + \frac{2a_1}{a_2 - \sqrt{a_2^2 + 4a_1}}L\right).$$

The necessary and sufficient conditions for **stationarity** are:

$$a_1 + a_2 < 1, \quad -a_1 + a_2 < 1, \quad -a_2 < 1.$$

Difference Stationary or Integrated Processes

If in an $AR(p)$ there exists one unit root (e.g. $\alpha_j = 1$), then the first difference is an $AR(p - 1)$:

$$\begin{aligned} a(L) &= (1 - \alpha_1 L) \cdots (1 - \alpha_{j-1} L)(1 - \alpha_{j+1} L) \cdots (1 - \alpha_p L)(1 - L) \\ &= (1 - b_1 L - \cdots - b_{p-1} L^{p-1})(1 - L) = b(L)(1 - L). \end{aligned}$$

If the remaining $p - 1$ roots are greater than one in absolute value, then the process is **difference stationary** or **integrated of order one**, denoted $I(1)$.

More generally, if an $AR(p)$ in y_t contains d unit roots, then $\Delta^d y_t$ is an $AR(p - d)$, and if all remaining roots are greater than one in absolute value, then the process is $I(d)$.

Dickey-Fuller Test

Dickey Fuller tests for unit roots in an AR(1), i.e. ($a_1 = 1$).

To implement the test, re-arrange the model as

$$y_t = a_1 y_{t-1} + \varepsilon_t,$$

$$y_t - y_{t-1} = a_1 y_{t-1} - y_{t-1} + \varepsilon_t,$$

$$\Delta y_t = \gamma y_{t-1} + \varepsilon_t,$$

where $\gamma = a_1 - 1$.

Test $H_0 : \gamma = 0$ against $H_1 : \gamma_1 < 0$, with critical values tabulated under the *Dickey-Fuller distribution*.

What about testing for unit roots in an AR(p)?

Augmented Dickey-Fuller Test

For the AR(2), follow a similar strategy to obtain

$$y_t = a_1 y_{t-1} + a_2 y_{t-2} + \varepsilon_t,$$

$$y_t - y_{t-1} = a_1 y_{t-1} - y_{t-1} + a_2 y_{t-2} + a_2 y_{t-1} - a_2 y_{t-1} + \varepsilon_t,$$

$$\Delta y_t = \gamma y_{t-1} + b_1 \Delta y_{t-1} + \varepsilon_t,$$

where $\gamma = a_1 + a_2 - 1$ and $b_1 = -a_2$.

In the AR(2), a unit root exists iff $a_1 + a_2 = 1$ or equivalently $\gamma = 0$.

For the general AR(p) case, estimate

$$\Delta y_t = \gamma y_{t-1} + b_1 \Delta y_{t-1} + \cdots + b_{p-1} \Delta y_{t-p+1} + \varepsilon_t,$$

and test $\gamma = 0$.

Including Deterministic Regressors and Trends

$$y_t = a_1 y_{t-1} + \cdots + a_p y_{t-p} + \varepsilon_t,$$

$$\Leftrightarrow \Delta y_t = \gamma y_{t-1} + b_1 \Delta y_{t-1} + \cdots + b_{p-1} \Delta y_{t-p+1} + \varepsilon_t,$$

$$y_t = a_0 + a_1 y_{t-1} + \cdots + a_p y_{t-p} + \varepsilon_t,$$

$$\Leftrightarrow \Delta y_t = a_0 + \gamma y_{t-1} + b_1 \Delta y_{t-1} + \cdots + b_{p-1} \Delta y_{t-p+1} + \varepsilon_t,$$

$$y_t = a_0 + \delta t + a_1 y_{t-1} + \cdots + a_p y_{t-p} + \varepsilon_t,$$

$$\Leftrightarrow \Delta y_t = a_0 + \delta + \gamma y_{t-1} + b_1 \Delta y_{t-1} + \cdots + b_{p-1} \Delta y_{t-p+1} + \varepsilon_t.$$

Augmented Dickey-Fuller Test

To implement the **Augmented Dickey-Fuller** (ADF), estimate the most general specification:

$$\Delta y_t = a_0 + \delta t + \gamma y_{t-1} + b_1 \Delta y_{t-1} + \cdots + b_{p-1} \Delta y_{t-p+1} + \varepsilon_t.$$

To test for a **unit root**, use $H_0 : \gamma = 0$ against $H_1 : \gamma < 0$. This is the τ_τ test, with critical values tabulated in Enders (Table A).

- interpretation: under H_0 , the process is $I(1)$ with a drift and deterministic trend (i.e., exhibits **linear growth**); under H_1 , it is trend stationary.

To test for both a **unit root and deterministic trend**, use $H_0 : \delta = 0, \gamma = 0$ against $H_1 : \delta \neq 0, \gamma < 0$. This is the ϕ_3 test, with critical values tabulated in Enders (Table B).

- interpretation: under H_0 , the process is $I(1)$ with a drift (i.e., exhibits **constant growth**); under H_1 , it is trend stationary.

Other Variants of the ADF

Omitting δt and testing $H_0 : \gamma = 0$ against $H_1 : \gamma < 0$ leads to the τ_μ test (critical values in Enders, Table A).

Omitting δt and testing $H_0 : a_0 = 0, \gamma = 0$ against $H_1 : a_0 \neq 0, \gamma < 0$ leads to the ϕ_1 test (critical values in Enders, Table B).

Omitting $a_0 + \delta t$ and testing $H_0 : \gamma = 0$ against $H_1 : \gamma < 0$ leads to the τ test (critical values in Enders, Table A).

All ADF test statistics have **non-standard** distributions.

ADF Test: Example 1

Null Hypothesis: LGDPPC has a unit root

Exogenous: Constant, Linear Trend

Lag Length: 1 (Automatic - based on SIC, maxlag=13)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-2.856187	0.1801
Test critical values:		
1% level	-4.022586	
5% level	-3.441111	
10% level	-3.145082	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation

Included observations: 145 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LGDPPC(-1)	-0.084277	0.029507	-2.856187	0.0049
D(LGDPPC(-1))	0.061183	0.081826	0.747723	0.4559
C	0.748960	0.261091	2.868583	0.0048
@TREND("1973Q3")	0.000416	0.000143	2.910159	0.0042
R-squared	0.057617	Mean dependent var		0.004185
Adjusted R-squared	0.037566	S.D. dependent var		0.009582
S.E. of regression	0.009400	Akaike info criterion		-6.468984
Sum squared resid	0.012459	Schwarz criterion		-6.386867
Log likelihood	473.0013	Hannan-Quinn criter.		-6.435617
F-statistic	2.873541	Durbin-Watson stat		1.987599
Prob(F-statistic)	0.038485			

ADF Test: Example 2

Null Hypothesis: LGDPPC has a unit root

Exogenous: Constant

Lag Length: 1 (Automatic - based on SIC, maxlag=13)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	0.217256	0.9729
Test critical values: 1% level	-3.475819	
5% level	-2.881400	
10% level	-2.577439	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation

Included observations: 145 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LGDPPC(-1)	0.000858	0.003950	0.217256	0.8283
D(LGDPPC(-1))	0.024860	0.082968	0.299638	0.7649
C	-0.003823	0.036359	-0.105155	0.9164
R-squared	0.001013	Mean dependent var		0.004185
Adjusted R-squared	-0.013057	S.D. dependent var		0.009582
S.E. of regression	0.009644	Akaike info criterion		-6.424448
Sum squared resid	0.013208	Schwarz criterion		-6.362860
Log likelihood	468.7725	Hannan-Quinn criter.		-6.399423
F-statistic	0.072011	Durbin-Watson stat		1.970315
Prob(F-statistic)	0.930555			

Summary

Many economic variables appear to be nonstationary.

- e.g., trending behaviour in real per capita GNP
- wandering behaviour in interest rates

The type of nonstationary process has important economic implications.

- **deterministic trends**: predictable behaviour with no long-run memory (shocks “die out”)
- **stochastic trends (& unit roots)**: trend in variance implying growing unpredictability in the future, permanent effect of shocks, complete long-run memory

Trend stationary vs. difference stationary processes.

- It is important for economic reasons to know if a process is TS or DS (e.g. trend-cycle decomposition); can test this using (Augmented) Dickey-Fuller.